VIT BHOPAL

FUNDAMENTALS IN AI AND ML

(CSA2001)

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Smart Network Bandwidth Monitor

PROBLEM STATEMENT

Users often experience slow internet but cannot identify which applications are consuming bandwidth. There is no simple, real-time, and lightweight tool to monitor network usage and visualize upload and download speeds. This project aims to develop a Smart Network Bandwidth Monitor to track real-time internet usage and analyze network performance efficiently.

SCOPE OF THE PROJECT

The Smart Network Bandwidth Monitor is designed to provide real-time monitoring and analysis of internet usage on a system. The project focuses on tracking upload and download speeds, visualizing bandwidth consumption, and helping users identify network performance issues. This tool aims to offer a lightweight, user-friendly solution for monitoring network activity and improving overall internet usage efficiency.

FUNCTIONAL REQUIREMENTS

1. **Real-Time Network Monitoring**
   The system should monitor and display upload and download speeds continuously in real time.
2. **Bandwidth Calculation**
   The application should calculate internet speed based on bytes sent and received over a period of time.
3. **Data Visualization**
   The system should generate graphs to represent upload and download speed trends for better understanding.
4. **User Interface Display**
   The application should provide a simple dashboard displaying network usage clearly.

5. **Continuous Data Collection**
   The system should continuously collect and update network usage data at regular intervals.

NON FUNCTIONAL REQUIREMENTS

1. **Performance**
   The system should operate efficiently without significantly affecting system performance.
2. **Usability**
   The interface should be simple, easy to use, and understandable for all users.
3. **Reliability**
   The system should run continuously and provide accurate network monitoring.
4. **Portability**
   The application should run on different operating systems such as Windows, Linux, and macOS.
5. **Maintainability**
   The code should be modular and easy to update for future enhancements.

SYSTEM ARCHITECTURE

Overview

The Smart Network Bandwidth Monitor follows a simple layered architecture where the system collects network data from the operating system, processes it using Python, and displays the results using a graphical interface. The architecture ensures real-time monitoring, efficient processing, and easy visualization of network usage.

The system consists of four main components:

- Data Collection Layer
- Processing Layer
- Visualization Layer

- User Interface Layer

SYSTEM ARCHITECTURE COMPONENTS

1. Data Collection Layer

This is the first layer of the system where network data is collected from the computer.

Functions

- Collect bytes sent and received
- Monitor network activity
- Fetch system network statistics

Tools Used

- Python
- psutil library

The `psutil` library interacts with the operating system and retrieves real-time network usage data.

2. Processing Layer

This layer processes the collected data and calculates network speed.

Functions

- Calculate upload speed
- Calculate download speed
- Convert bytes to KB/s or MB/s
- Store processed data temporarily

Working

The system calculates:

Upload Speed = Bytes Sent Difference / Time

Download Speed = Bytes Received Difference / Time

This allows the system to calculate real-time internet speed.

3. Visualization Layer

This layer converts processed data into graphs and charts.

Functions

- Generate real-time graphs
- Show bandwidth usage trends
- Update graphs dynamically

Tools Used

- Matplotlib
- Pandas

The system continuously updates graphs showing upload and download speeds.

4. User Interface Layer

This is the front-end layer where users interact with the system.

Functions

- Display upload speed
- Display download speed
- Show real-time graphs
- Provide dashboard view

Tools Used

- Streamlit

Streamlit creates a simple web-based dashboard for monitoring.

SYSTEM ARCHITECTURE FLOW

Operating System

↓

Data Collection (psutil)

↓

Processing Layer (Python)

↓

Data Storage (Pandas)

↓

Visualization (Matplotlib)

↓

User Interface (Streamlit Dashboard)

ARCHITECTURE DIAGRAM

```
+----------------------+
|   User Interface    |
|     (Streamlit)     |
+----------↑-----------+
          |
```

```
+----------|-----------+
|   Visualization     |
|    (Matplotlib)     |
+----------↑-----------+
          |
+----------|-----------+
|   Processing Layer   |
|     (Python)        |
+----------↑-----------+
          |
+----------|-----------+
|  Data Collection     |
|     (psutil)        |
+----------↑-----------+
          |
+----------------------+
|   Operating System   |
+----------------------+
```

USE CASE DIAGRAM(TEXTUAL)

```
                +---------------------------+
                |   Smart Network Monitor   |
                +---------------------------+

      User
       |
       |
       |------------------> Start Monitoring
       |
       |------------------> View Upload Speed
       |
       |------------------> View Download Speed
       |
       |------------------> View Real-Time Graph
       |
       |------------------> Analyze Bandwidth Usage
       |
       |------------------> Stop Monitoring
```

WORKFLOW/ACTIVITY DIAGRAM

```
Start
  ↓
User Runs Network Monitor
  ↓
System Initializes Libraries
  ↓
Collect Initial Network Data
  ↓
Start Monitoring Loop
  ↓
Collect Network Usage Data
  ↓
Calculate Upload Speed
  ↓
Calculate Download Speed
  ↓
Store Data Temporarily
  ↓
Generate Graphs
  ↓
Display Results on Dashboard
```

```
  ↓
Update Data Continuously
  ↓
User Stops Monitoring?
     ↓            ↓
    No           Yes
     ↓            ↓
Continue Loop    Stop Monitoring
     ↓            ↓
     ←-------------
           ↓
         End
```

SEQUENCE DIAGRAM

```
User          UI (Streamlit)       Network Monitor        System (psutil)        Graph Modu
 |                 |                     |                     |                    |
 |----Start App---->|                     |                     |                    |
 |                 |----Initialize------>|                     |                    |
 |                 |                     |----Request Data---->|                    |
 |                 |                     |<---Network Data-----|                    |
 |                 |                     |----Process Data-----|                    |
 |                 |<---Display Speed----|                     |                    |
 |                 |                     |----Send Data--------------------------->|
 |                 |<---Display Graph--------------------------------------------|
 |                 |                     |                     |                    |
 |<---View Output---|                     |                     |                    |
 |                 |                     |                     |                    |
 |----Stop----------|                     |                     |                    |
 |                 |----Terminate------->|                     |                    |
```

ER DIAGRAM

```
+----------------------+
|      User            |
+----------------------+
| - start()            |
| - stop()             |
+----------+-----------+
           |
           v

+----------------------+
|  NetworkMonitor      |
+----------------------+
| - upload_speed       |
| - download_speed     |
| - data               |
+----------------------+
| + get_network_data() |
| + calculate_speed()  |
| + update_data()      |
+----------+-----------+
           |
```

```
            v

+----------------------+
|  DataVisualizer      |
+----------------------+
| - graph_data         |
+----------------------+
| + generate_graph()   |
| + update_graph()     |
+----------+-----------+
           |
           v

+----------------------+
| StreamlitInterface   |
+----------------------+
| - display_data       |
+----------------------+
| + show_dashboard()   |
| + update_display()   |
+----------------------+
```

DESIGN DECISION AND RATIONAL

Design Decisions (Point-wise)

1. Python as Programming Language

Decision: Use Python for development
Rationale: Easy to implement, strong library support, and suitable for real-time monitoring.

2. Use of psutil Library

Decision: Use psutil for collecting network data
Rationale: Provides accurate and real-time system network statistics.

3. Streamlit for User Interface

Decision: Use Streamlit for dashboard creation
Rationale: Simple UI development and real-time display capability.

4. Real-Time Monitoring

Decision: Implement continuous monitoring
Rationale: Helps users detect network issues instantly.

5. Data Visualization using Matplotlib

Decision: Use Matplotlib for graphs
Rationale: Simple and effective visualization of bandwidth usage.

6. Modular Architecture

Decision: Separate data collection, processing, and visualization
Rationale: Improves maintainability and scalability.

7. Lightweight Design

Decision: Keep system simple without database
Rationale: Faster performance and easy implementation.

IMPLEMENTATION DETAILS

- Python used for development
- psutil for network data collection
- Calculate upload and download speed
- Pandas for temporary data storage
- Matplotlib for graph visualization
- Streamlit for user interface
- Real-time monitoring using loop
- Display speed and graphs continuously

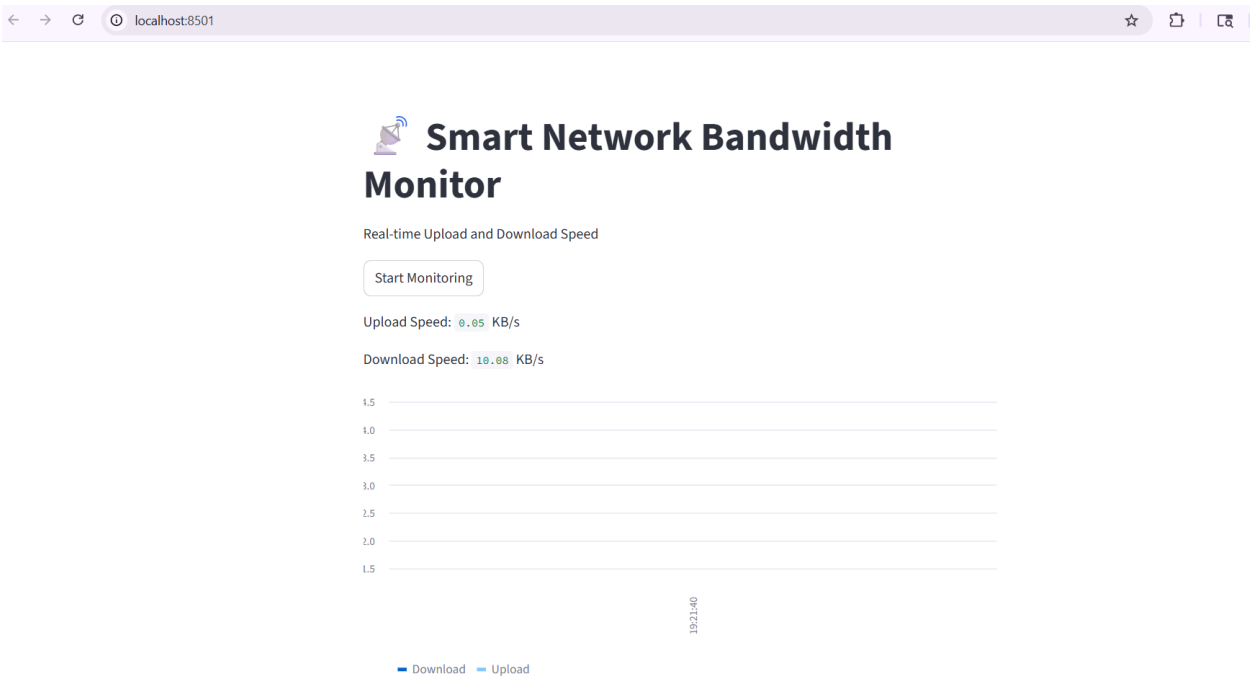
SOURCE CODE

> Users > SHRESTHA SARKAR > OneDrive > Desktop > network_monitor.py > ...

```python
import streamlit as st
import psutil
import pandas as pd
import time
from datetime import datetime

st.title("📡 Smart Network Bandwidth Monitor")

st.write("Real-time Upload and Download Speed")

# Create dataframe
data = pd.DataFrame(columns=["Time", "Upload", "Download"])

# Get initial values
old_value = psutil.net_io_counters()

# Button to start monitoring
if st.button("Start Monitoring"):

    for i in range(50):   # runs 50 times

        time.sleep(1)

        new_value = psutil.net_io_counters()

        upload = (new_value.bytes_sent - old_value.bytes_sent) / 1024
        download = (new_value.bytes_recv - old_value.bytes_recv) / 1024

        old_value = new_value

        current_time = datetime.now().strftime("%H:%M:%S")

        new_row = {
            "Time": current_time,
            "Upload": upload,
```

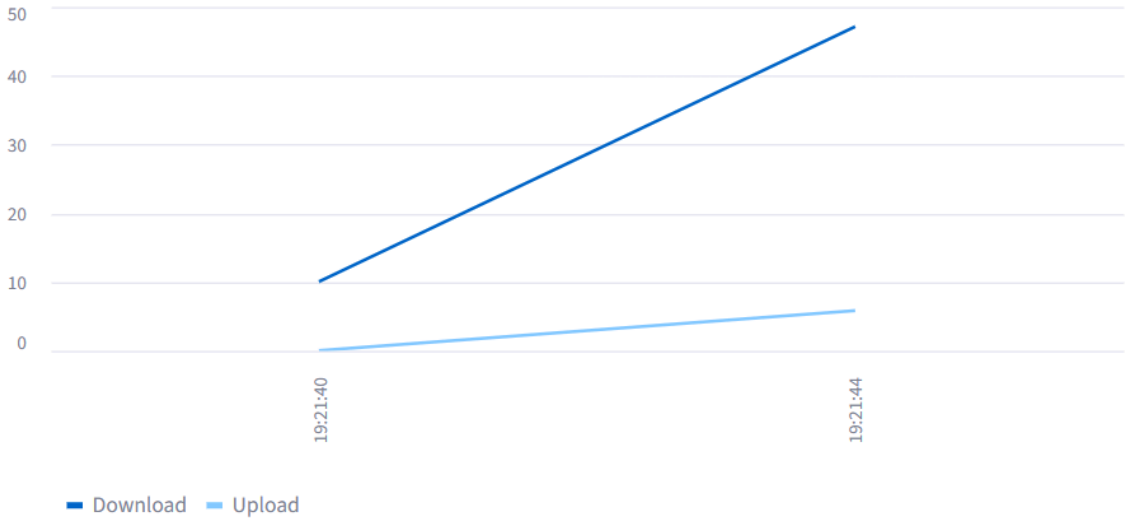
```
            "Download": download
        }

        data = pd.concat([data, pd.DataFrame([new_row])], ignore_index=True)

        st.write("Upload Speed:", round(upload,2), "KB/s")
        st.write("Download Speed:", round(download,2), "KB/s")

        st.line_chart(data.set_index("Time"))

```

OUTPUT



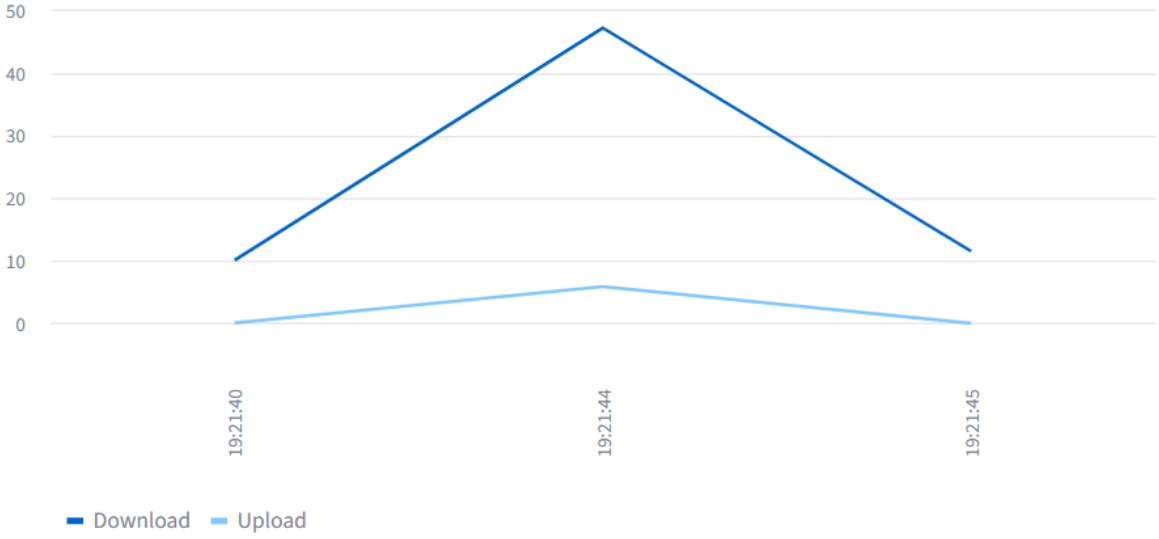
Upload Speed: `5.86` KB/s

Download Speed: `47.15` KB/s



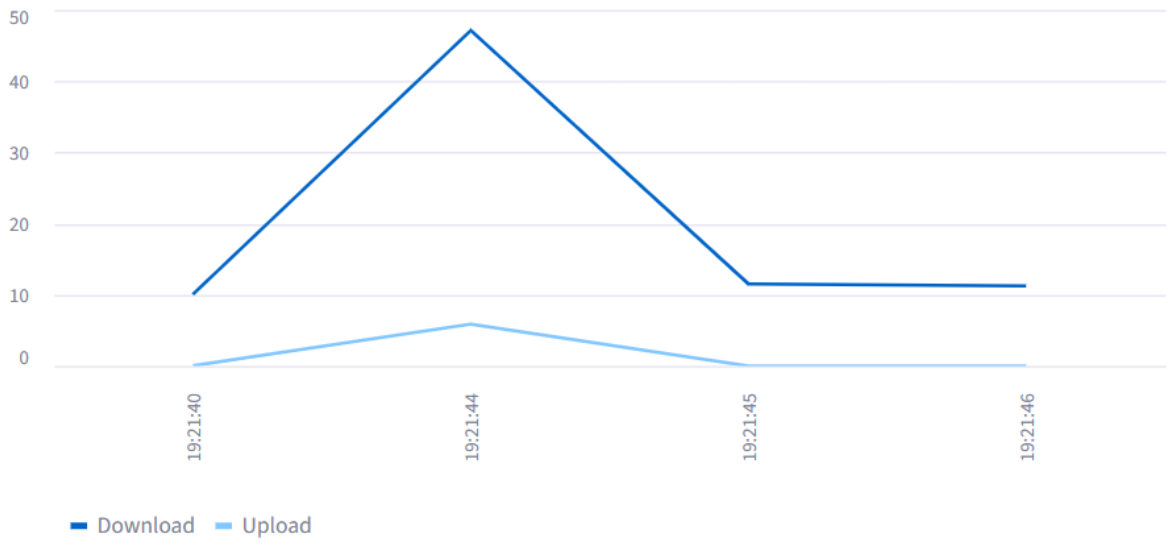
Upload Speed: `0.0` KB/s

Download Speed: `11.53` KB/s



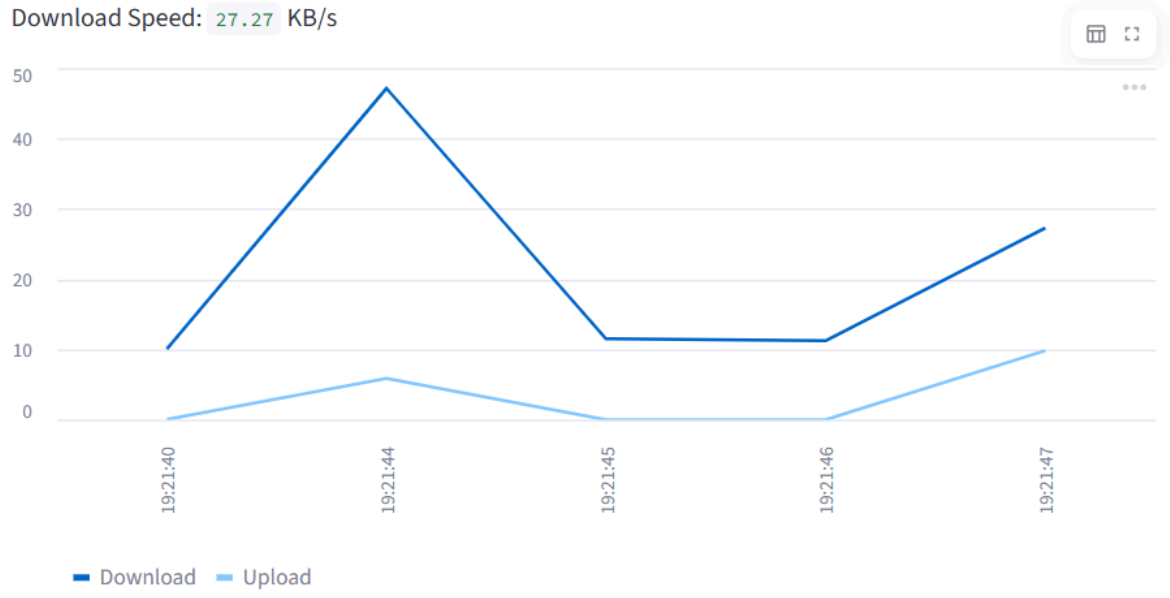
Upload Speed: `0.0` KB/s

Download Speed: `11.25` KB/s



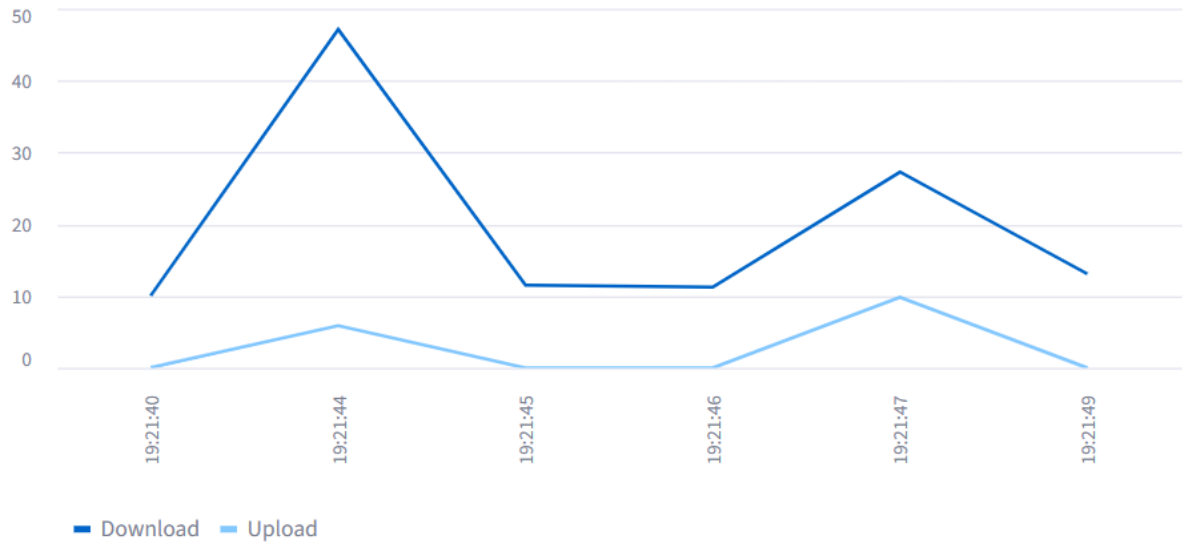
Upload Speed: `9.84` KB/s

Download Speed: `27.27` KB/s



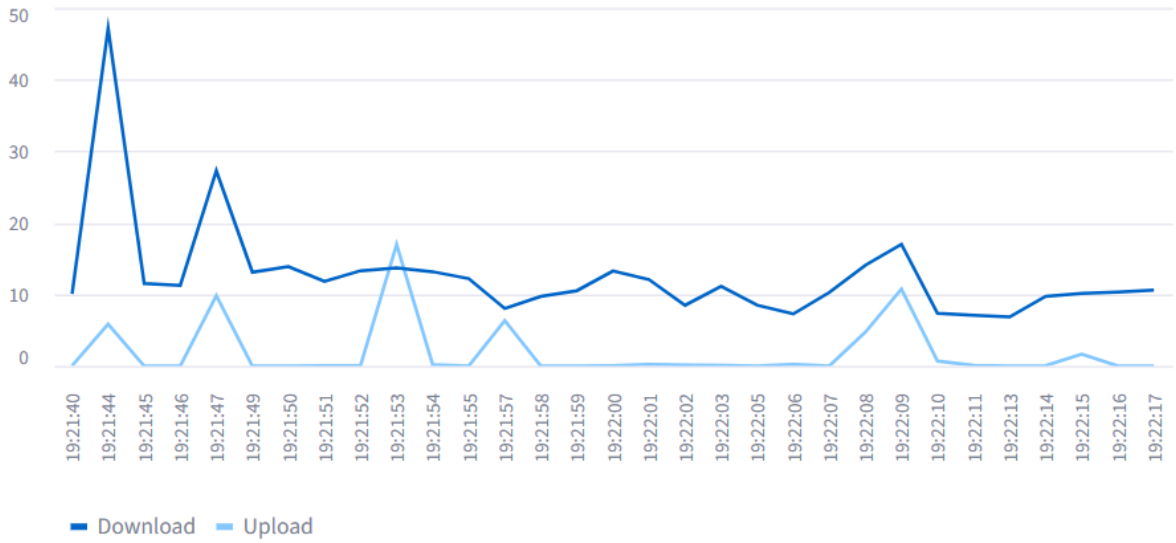
Upload Speed: `0.0` KB/s

Download Speed: `13.1` KB/s



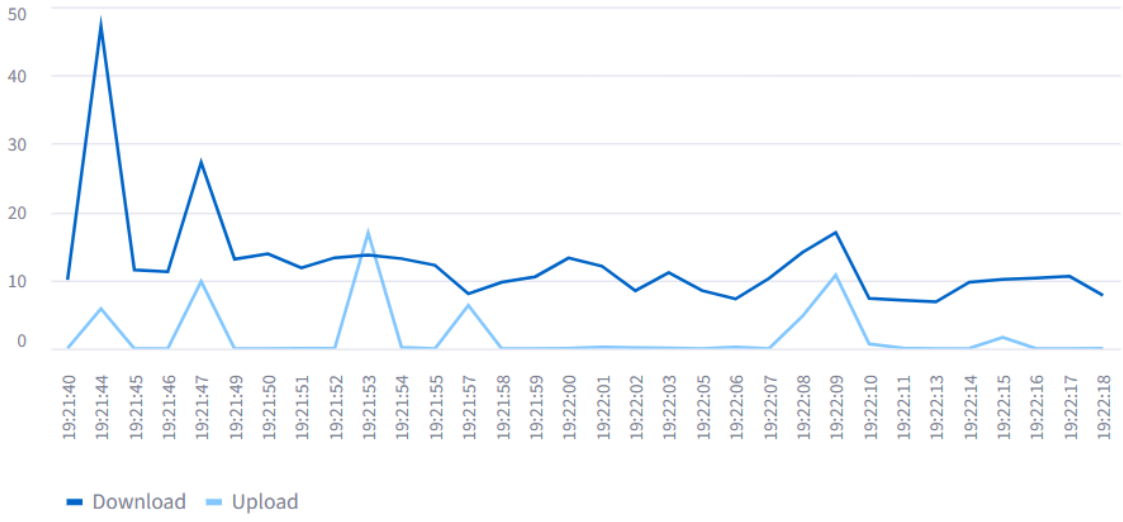
Upload Speed: `0.0` KB/s

Download Speed: `10.59` KB/s



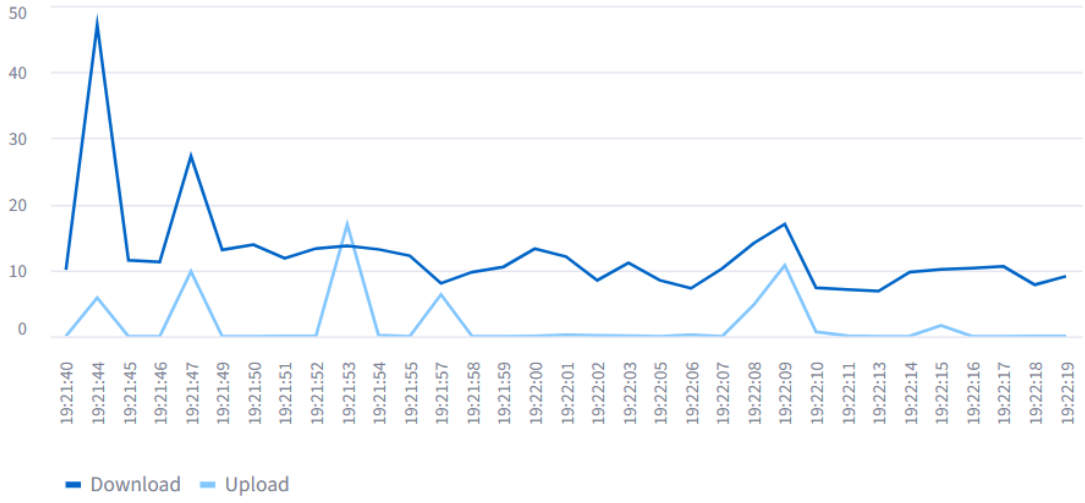
Upload Speed: `0.05` KB/s

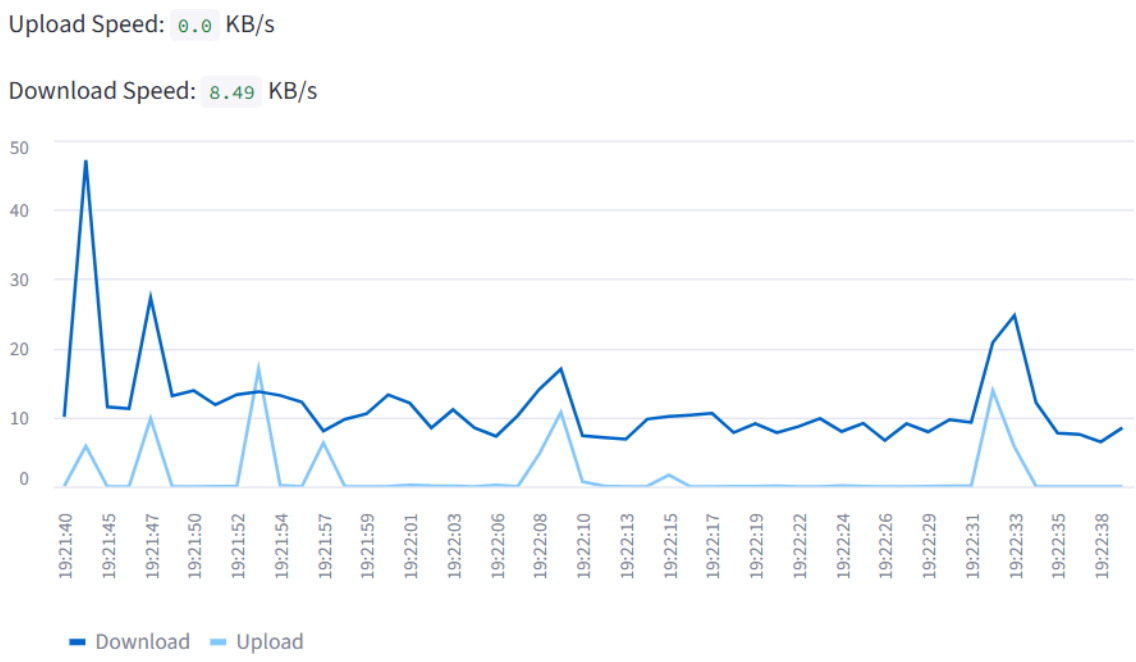
Download Speed: `7.81` KB/s



Upload Speed: `0.05` KB/s

Download Speed: `9.1` KB/s





TESTING APPROACH

- **Unit Testing** — Tested individual modules like data collection and speed calculation
- **Integration Testing** — Verified interaction between monitoring, processing, and UI
- **Functional Testing** — Checked real-time upload and download speed display
- **Performance Testing** — Ensured system runs smoothly without slowing the computer
- **User Interface Testing** — Verified dashboard and graphs display correctly
- **Real-Time Testing** — Tested monitoring with active internet usage
- **Error Testing** — Checked behavior during no internet or low connectivity

CHALLENGES FACED

- Difficulty in collecting real-time network data
- Handling continuous data updates efficiently
- Calculating accurate upload and download speeds
- Managing real-time graph updates
- Ensuring system performance is not affected
- Integrating multiple Python libraries
- Handling no-internet or low-connectivity situations
- Creating a simple and user-friendly interface

LEARNINGS AND KEY TAKEAWAYS

- Mastered file handling (open, read, write, append)
- Understood persistent storage using text files
- Implemented basic user authentication
- Learned menu-driven program design
- Explored timestamp logging with datetime
- Realized importance of input validation & password security
- Gained confidence in building complete console apps from scratch

FUTURE ENHANCEMENTS

- App-wise bandwidth usage monitoring
- Daily and weekly usage reports
- Data usage alerts and notifications
- Bandwidth usage prediction
- Export data to CSV or Excel
- Advanced analytics dashboard
- Multi-device network monitoring
- Mobile-friendly interface

REFERENCES

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https://www.python.org/doc/

Psutil Documentation
https://psutil.readthedocs.io/

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https://docs.streamlit.io/

Pandas Documentation
https://pandas.pydata.org/docs/

Matplotlib Documentation
https://matplotlib.org/stable/index.html

Python Networking Concepts
https://realpython.com/python-networking/

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https://docs.streamlit.io/library/get-started

Psutil GitHub Repository
https://github.com/giampaolo/psutil